ENGLISH COMMUNICATION PRACTICE BOT

1. Project Title

English Communication Practice Bot

An AI-powered chatbot designed to help users improve English speaking and communication skills through real-time conversation and grammar feedback.

2. Project Overview

The English Communication Practice Bot is an AI-driven conversational system built using Natural Language Processing (NLP) and Large Language Models (LLMs).

The system allows users to:

- Practice English conversations
- Receive grammar correction
- Improve vocabulary
- Store and review previous chats
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3. Objectives

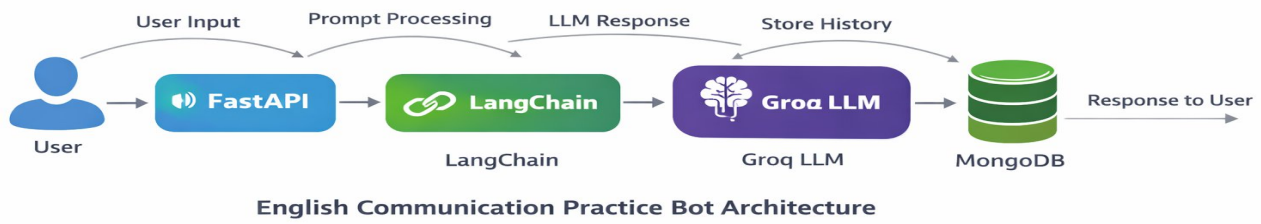
The main objectives of this project are:

- To develop an AI-based English conversation assistant
- To provide real-time grammar correction
- To store user conversation history
- To implement memory-based contextual responses
- To build scalable backend APIs using FastAPI
- To deploy the system for real-time access

4. Technologies Used

- Python
- FastAPI
- Groq API
- LangChain
- MongoDB
- Render(Deployment)

5. System Architecture



Architecture Explanation:

1. User sends message through client
2. FastAPI receives API request
3. LangChain formats structured prompt
4. Groq LLM generates intelligent response
5. MongoDB stores conversation history
6. Response is returned to user

6. Memory Implementation

Memory is implemented using **MongoDB database**.

Conversation history is stored in MongoDB. Each message contains user_id, message, response, and timestamp. The system retrieves the last few messages before generating a new response to maintain conversation context.

7. API Endpoints

POST /chat

Example Request :

```
{
    "user_id": "123",
    "message": "Hello"
}
```

Example Response:

```
{
  "response": "Hello! How can I help you practice English today?"
}
```

8. Deployment

The application is deployed using the Render cloud platform. Render automatically builds and deploys the application from a GitHub repository.

9. Results

The chatbot successfully generates responses for user inputs, stores chat history, and provides interactive English conversation practice. The API responds quickly and can support multiple users.

10. Conclusion

The English Communication Practice Bot demonstrates how Generative AI and NLP can be used to improve language learning. The integration of FastAPI, LangChain, Groq LLM, and MongoDB enables a scalable and intelligent conversational system